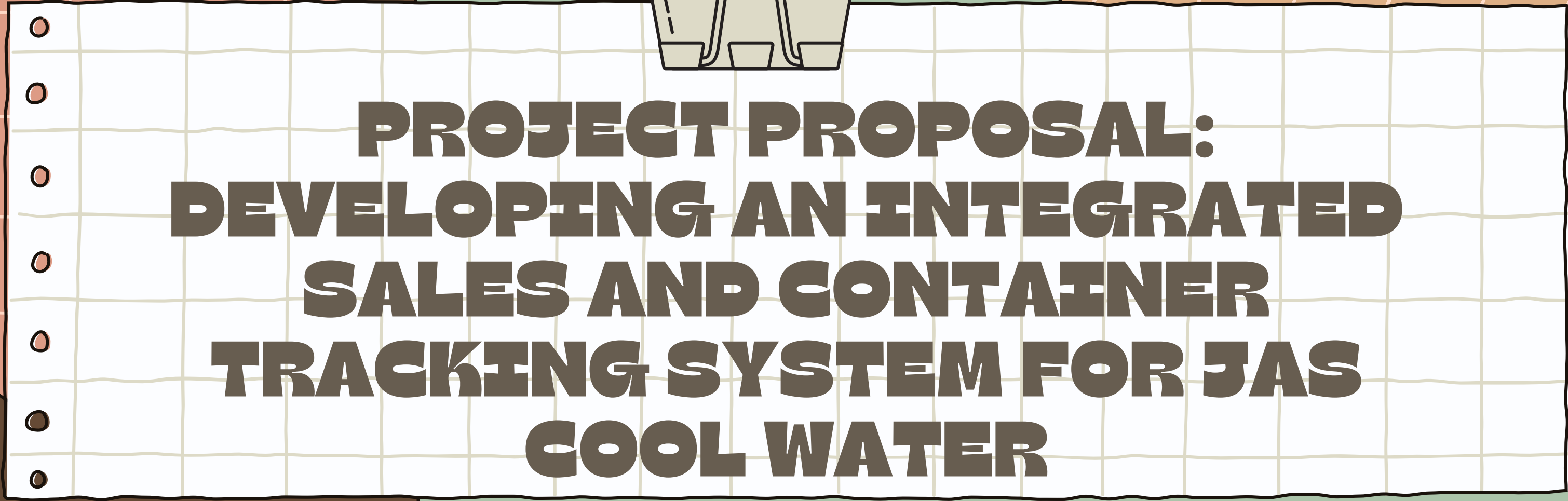


A white rectangular card with a light green grid pattern and a black border. On the left side, there are five small circles representing binder holes. The card is placed on a background with a green field, a blue sky, and a brown ground area. A yellow pencil is visible at the top center, and a yellow ruler is on the right side.

PROJECT PROPOSAL: DEVELOPING AN INTEGRATED SALES AND CONTAINER TRACKING SYSTEM FOR JAS COOL WATER

Presented by : Group 10

Dimla, Zeny Rose
Macaslang, Nepthalie Jezer
Magno, Joshua
Tejero, Harrys

INTRODUCTION - TRENDS

An effective information system is essential in supporting operational efficiency and management in the water refilling industry:

- Enhances accuracy and productivity through automated record-keeping
- Supports real-time decision-making and coordination among staff
- Ensures transparency and accountability in monitoring basyo circulation and returns
- Streamlines data access and retrieval
- Strengthens data security
- Provides actionable insights

The system serves as an integrated digital solution that automates

- Sales Transactions
- Monitor Inventory
- Record Basyo Movement/Circulation

INTRODUCTION - TRENDS

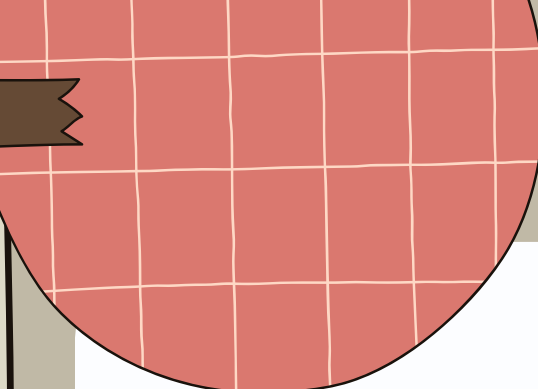
Technological and Digital Transformation in the Water Refilling Industry

- Computerized systems for sales and inventory (Bucaro et al., 2020)
- Integrated multiple features data systems for reduced errors (Norozpour et al., 2021)
- IoT-based monitoring for real-time inflow/outflow tracking to track stock visibility (Sourideth et al., 2024)

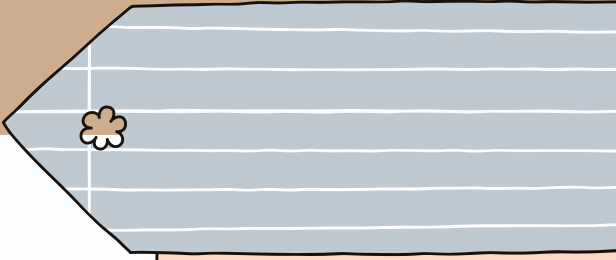
INTRODUCTION - ISSUES

Operational Concerns in Local Water Stations (e.g., Jas Cool Water)

- Uses a manual process for recording sales and tracking basyo/containers
- Frequently loses or miscounts basyo due to lack of systematic tracking
- Difficulty identifying which customers have returned or not returned their basyo
- Time-consuming and prone to human error, as most processes are done on paper
- Inaccurate sales and inventory records leading to untraceable asset loss.



INTRODUCTION - ISSUES



RESEARCH GAP

Designing a Computer-Based Inventory System of Crystal Clear Water Refilling Station
(Bucaro et al., 2020)

Functions

Their system automated sales and inventory recording through digital forms, helping reduce manual errors and improve data organization.

Gap:

However, its scope was limited to recording transactions. It lacked centralized monitoring capabilities, instant data retrieval, and asset protection features, specifically the ability to distinguish between Store-Owned and Customer-Owned containers, which is vital for preventing asset loss in a manual-heavy business.

INTRODUCTION - OBJECTIVES

General Objective: To design and develop a Sales and Container Tracking System for Jas Cool Water that transitions the business from a manual workflow to a centralized digital operation.

Key Project Goals:

- Automate sales recording and container tracking to eliminate manual errors.
- Centralize data management to ensure record security and reliability.
- Improve operational efficiency through real-time monitoring of assets.

INTRODUCTION - CONTRIBUTION

This project aligns with the **United Nations Sustainable Development Goals (SDGs)** — particularly:



SDG 9: Industry, Innovation, and Infrastructure

- Encourages digital innovation in small-scale businesses through automation and real-time data tracking.
- Strengthens business infrastructure with reliable and efficient record-keeping systems.



SDG 17: Partnerships for the Goals

- Fosters collaboration between local enterprises, communities, and technology developers.
- Demonstrates how small businesses can adopt technological solutions to contribute to sustainable growth.

PROJECT CONTEXT

Jas Cool Water

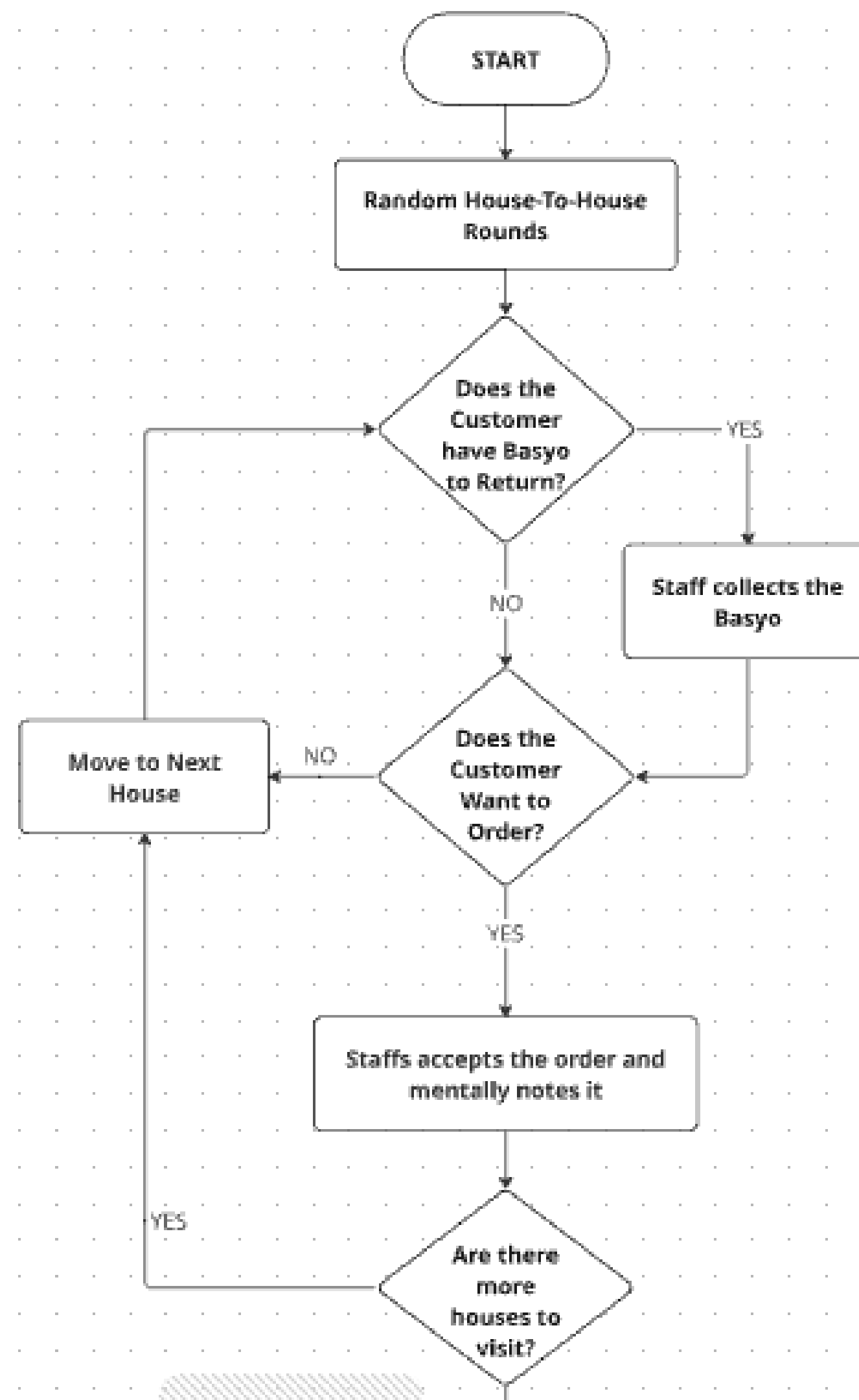
Location: Brgy. Perez, Bulakan, Bulacan, Philippines

Established: 2011, Owned by San Jose Family

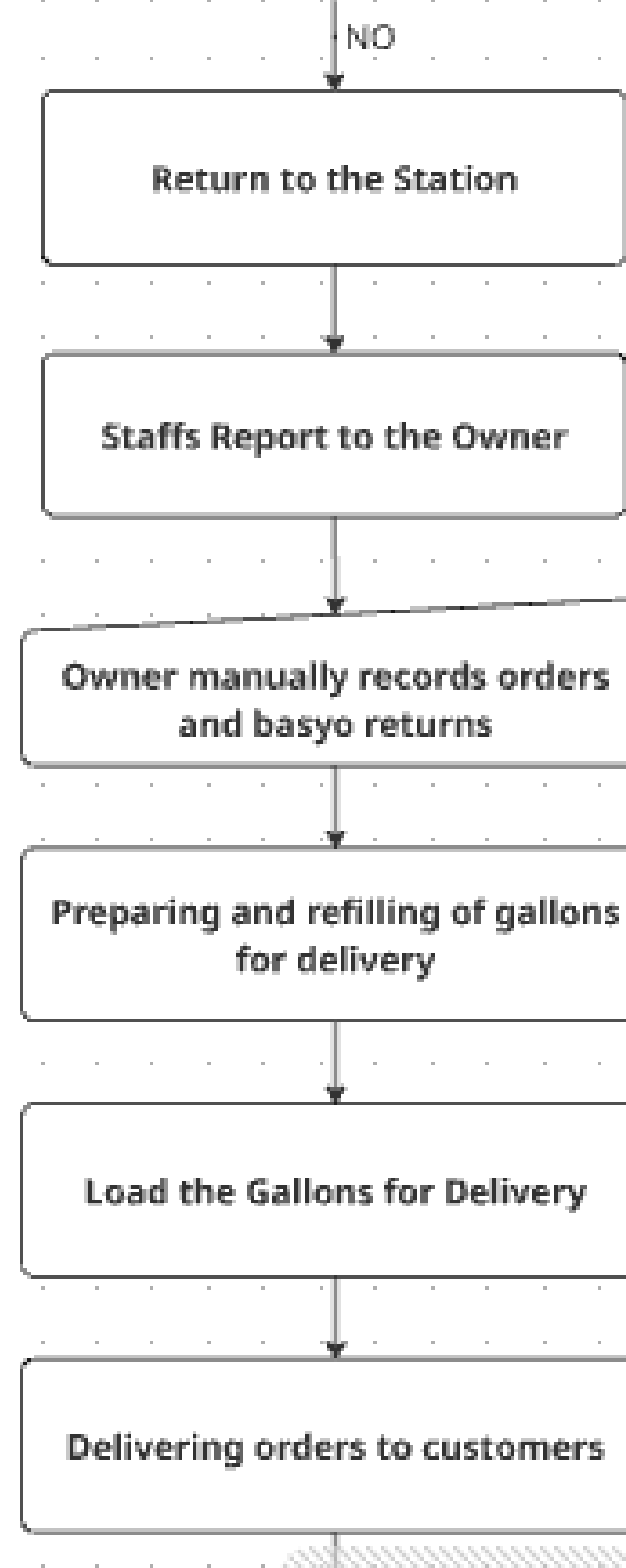
Services: Purified drinking water delivery, Walk-in customer service, Scheduled and recurring water delivery for households and small businesses, quality assurance through regular water testing and sanitation checks.

Promotion: Social media, Word of mouth, Local flyers and posters, House-to-house marketing

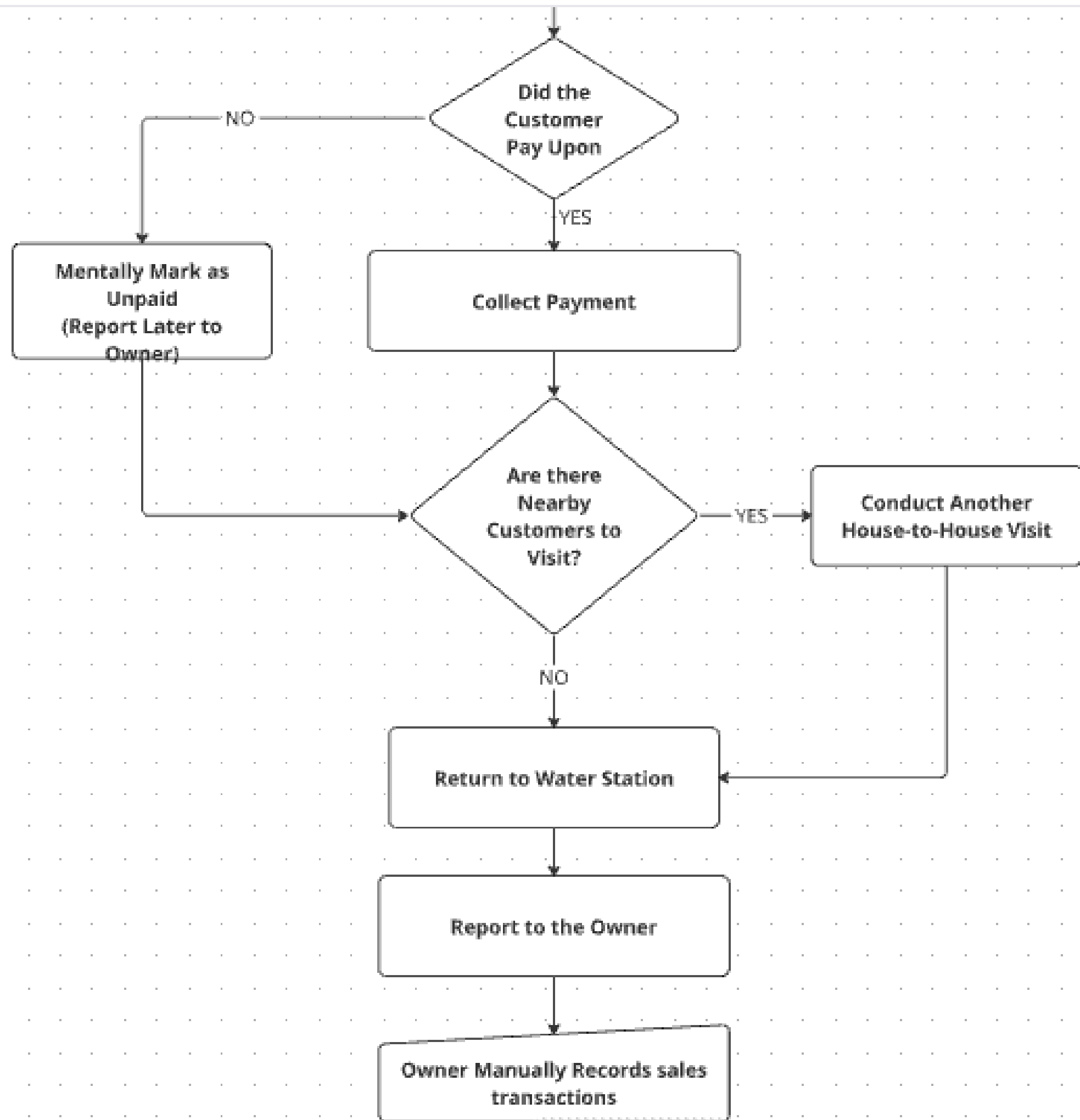
AS-IS FLOWCHART



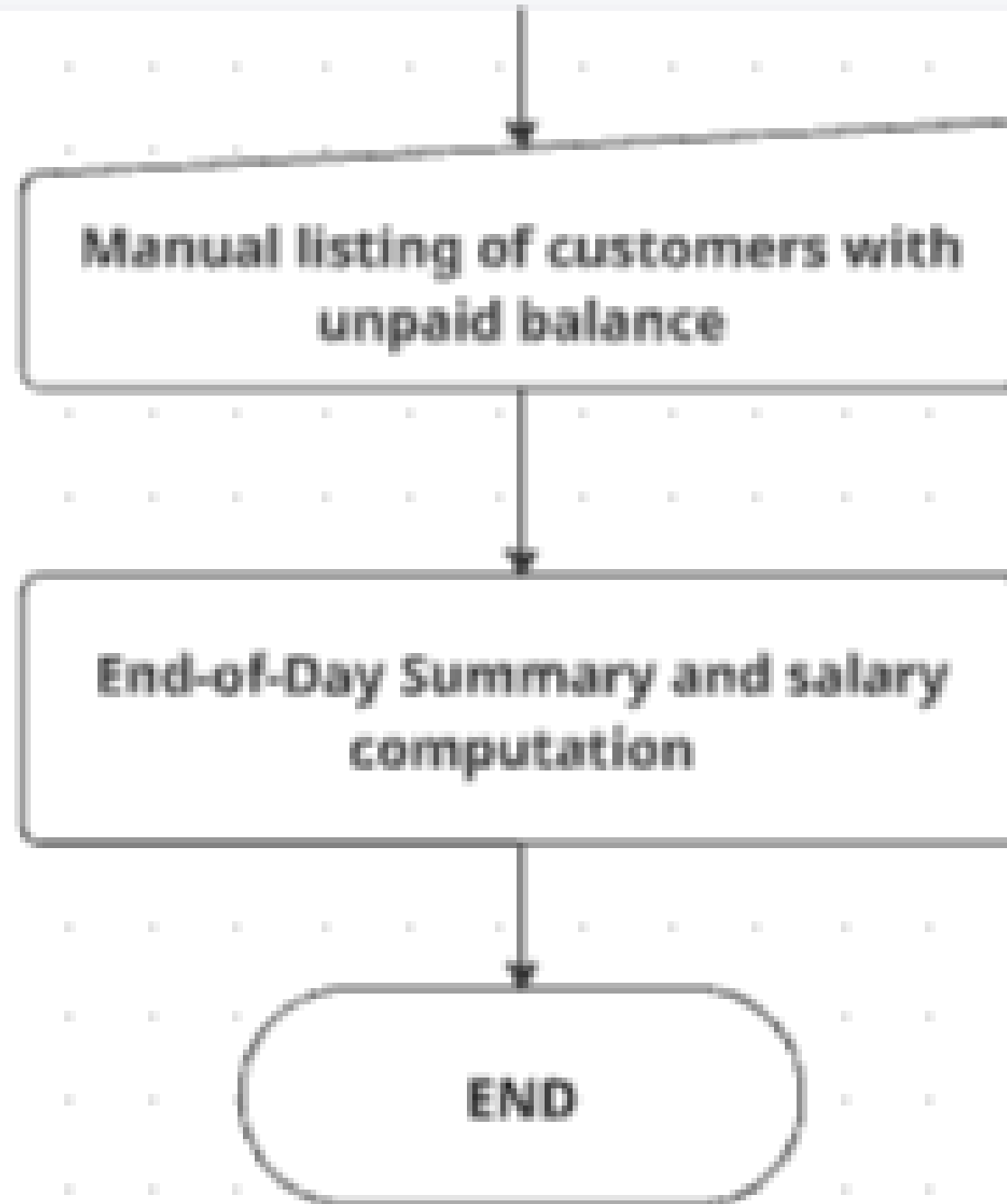
AS-IS FLOWCHART

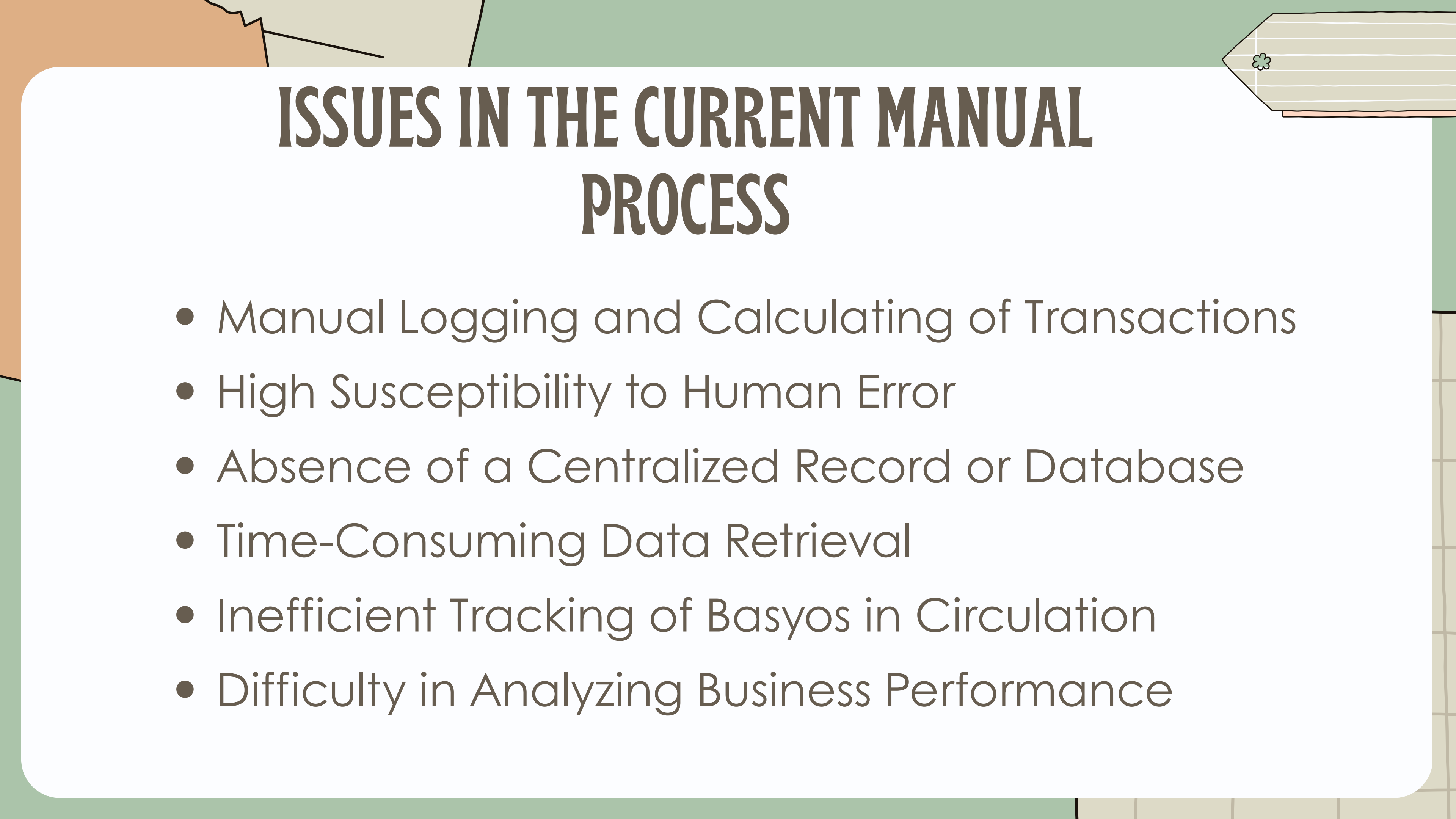


AS-IS FLOWCHART



AS-IS FLOWCHART





ISSUES IN THE CURRENT MANUAL PROCESS

- Manual Logging and Calculating of Transactions
- High Susceptibility to Human Error
- Absence of a Centralized Record or Database
- Time-Consuming Data Retrieval
- Inefficient Tracking of Basyos in Circulation
- Difficulty in Analyzing Business Performance

SUGGESTIONS/RECOMMENDATION

The owner and staffs recommends the development of an information system that can:

- Record customer profiles.
- Record transactions accurately.
- Track the basyo balance per customer to prevent shortages and disputes.
- Provide an option to record payment status.
- Quick and easy access to past transactions.
- Generate daily sales reports to support decision-making and review business performance.

STATEMENT OF THE PROBLEM

General Problem:

How can a Sales and Container Tracking System be designed and developed for Jas Cool Water in order to improve the operational performance and monitoring capabilities of its current manual process?

- 1.What is the current status of the manual sales and container tracking process of Jas Cool Water?
- 2. What operational issues and inefficiencies are present in the existing manual process?
- 3.What system features are required to address the owner's recommendations regarding **Sales Management, Container Tracking, and Automated Reporting**?
- 4.How can the proposed system be designed to integrate **Automated Tracking Algorithm** and **Centralized Database Management**?
- 5.How does the developed system improve the current manual process in terms of **Transaction Accuracy, Monitoring Efficiency, and Data Retrieval Capabilities**?

OBJECTIVES

General Objective:

To develop a Sales and Container Tracking System for Jas Cool Water that improves the operational performance and monitoring capabilities of the current manual process.

Assess the current status of Jas Cool Water's manual sales and container tracking workflow.

Identify the operational issues, inefficiencies, and risks associated with the business's existing manual processes.

Determine the system features required based on the owner's recommendations for **Sales Management, Container Tracking, and Automated Reporting**.

Design and develop the system utilizing **Automated Tracking Algorithm** and **Centralized Database Management**.

Evaluate the developed system's effectiveness in terms of **Transaction Accuracy, Monitoring Efficiency, and Data Retrieval Capabilities**.

SYSTEM OVERVIEW



The proposed solution is designed as a Role-Based Desktop Application featuring two distinct interfaces: a secure Admin Dashboard for management control and a streamlined Cashier Dashboard for daily transaction processing.

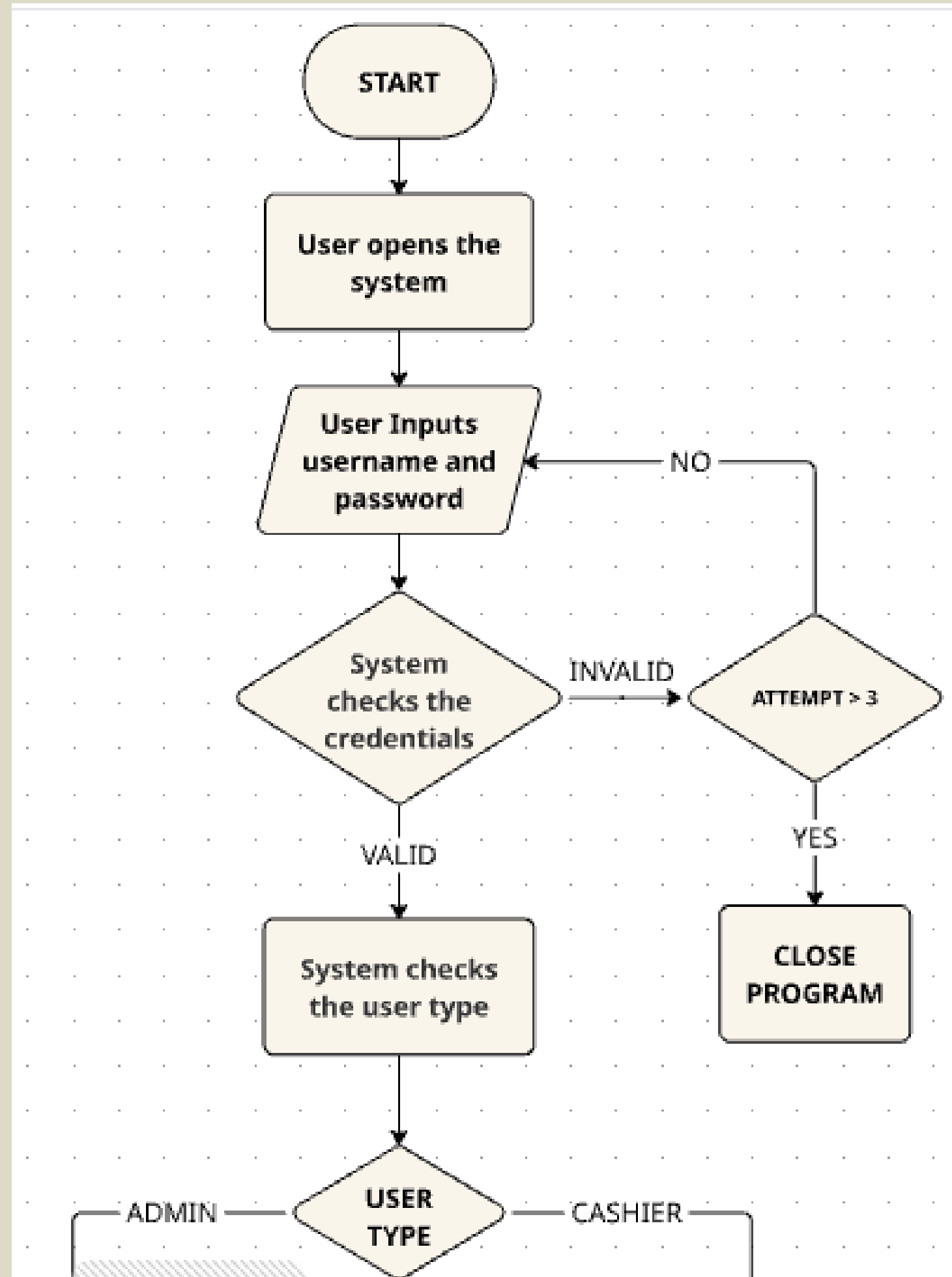
Admin Dashboard

- User Management Module
- Customer Management Module
- Basyo Tracking Module (Store vs. Customer Assets)
- Sales Report Module (Automated & Aging Reports)
- Inventory Module
- System Maintenance Module (Audit Logs & Backups)

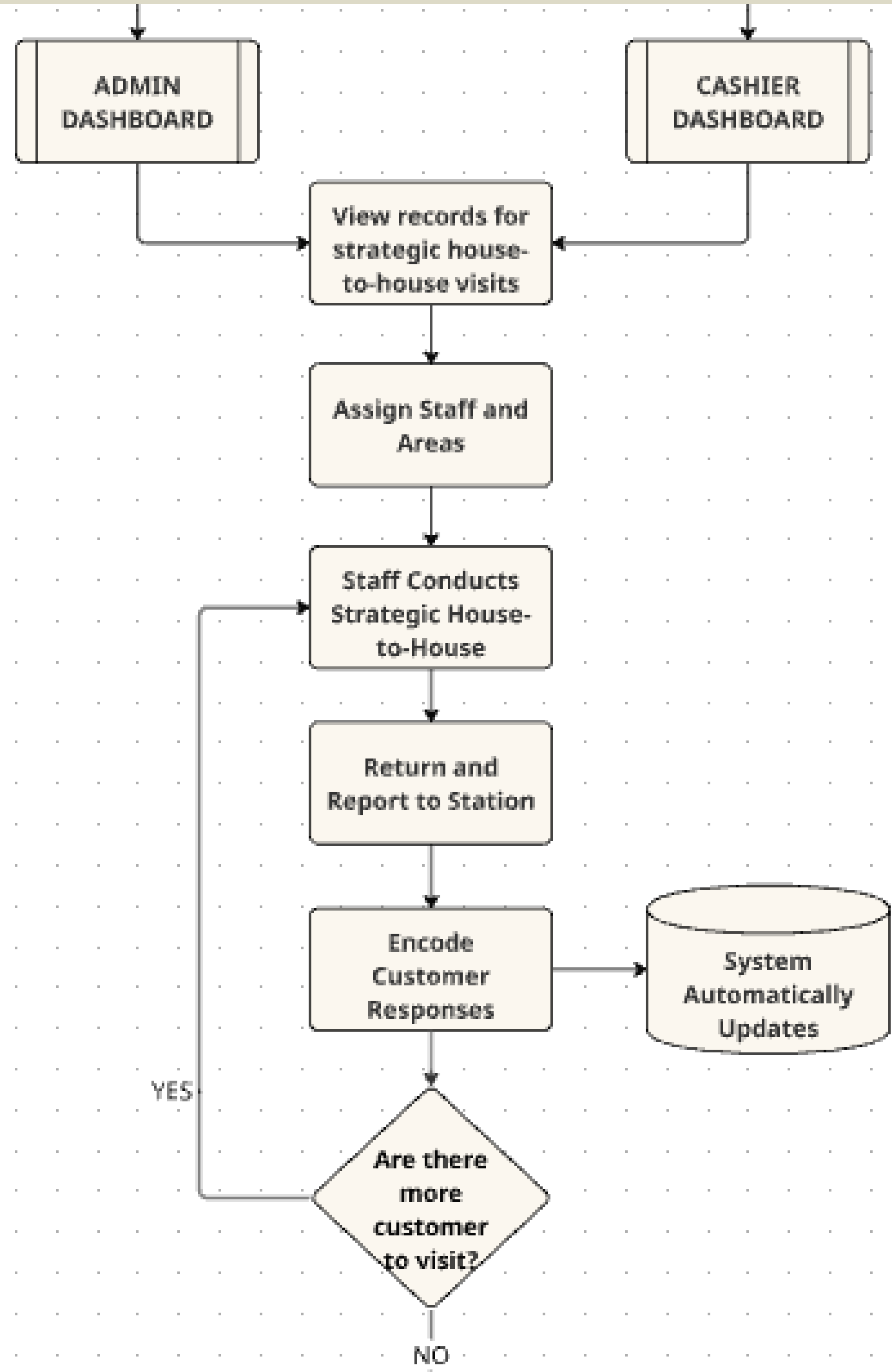
Cashier Dashboard

- Customer Management Module
- Sales Management Module
- Collection Module
- Basyo Balance View Module
- View Sales Module

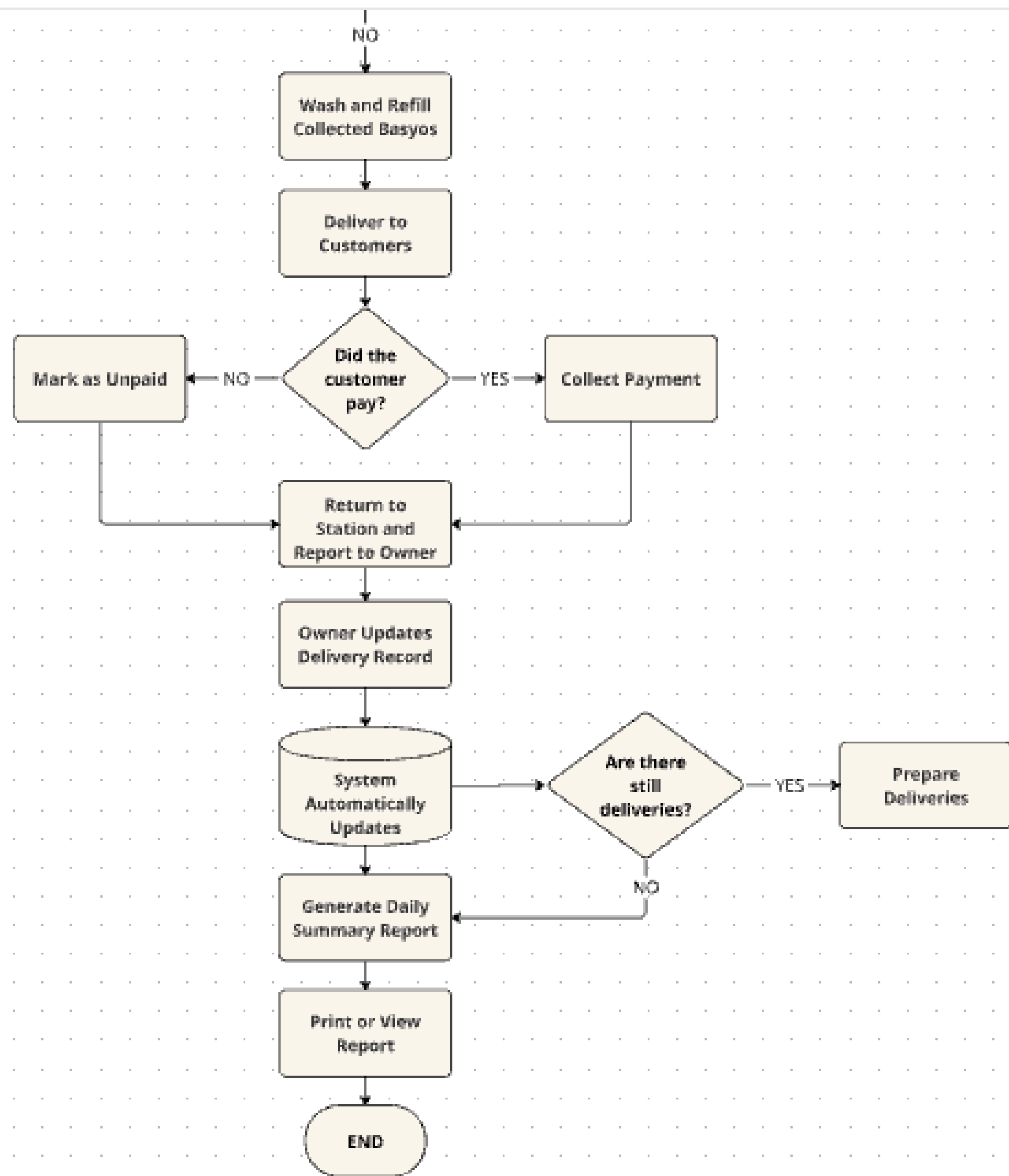
TO-BE PROCESS



- User Management Module
- Customer Management Module
- Container Tracking Module
- Sales Report Module
- Inventory Module
- System Maintenance Module



- Sales Management Module
- Collection Module
- Basyo Balance View Module

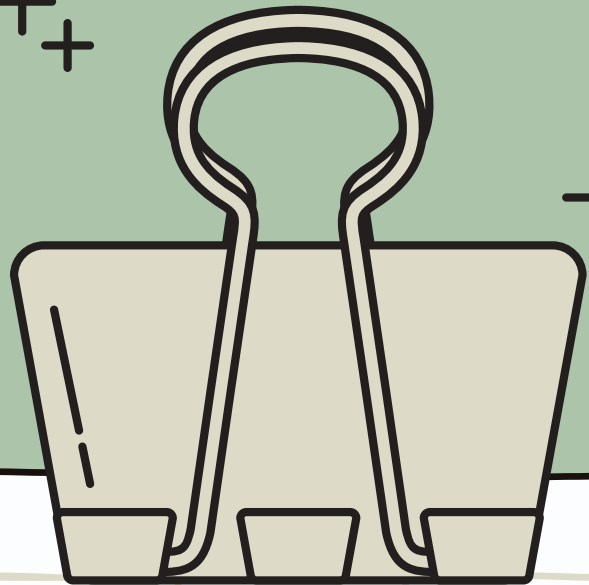


COMPARISON ON THE AS-IS AND TO-BE PROCESS

PROCESS	CURRENT (AS-IS)	IDENTIFIED PROBLEMS	PROPOSED SYSTEM SOLUTION (TO-BE)
Transaction Recording & Processing	Transactions and totals are manually written and computed in a notebook.	Records are often disorganized, illegible, and prone to manual calculation errors.	Transactions are encoded digitally with automated computations, ensuring records are organized, legible, searchable, and accurate.
Basyo Tracking	Owner relies on rough estimates or memory to know which customers still have basyos.	No clear or real-time record of basyo distribution and returns.	System provides Real-Time Monitoring Efficiency, instantly showing exactly which customer holds store assets.
Data Integrity & Security	Records are written in pencil/pen and can be altered or erased without a trace.	Susceptible to data manipulation and unauthorized changes by staff.	System enforces strict Audit Trails and System Logs, recording every edit or reversal to prevent manipulation.

PROCESS	CURRENT (AS-IS)	IDENTIFIED PROBLEMS	PROPOSED SYSTEM SOLUTION (TO-BE)
Inventory Management	Owner manually counts available empty basyos and bottles, mixed with the damaged ones.	Time-consuming and sometimes inaccurate	Inventory Module updates basyo quantities automatically whenever returns or orders are recorded in the system.
End-of-Day Summary	Owner manually totals daily sales and basyo counts using the notebook records.	Repetitive manual calculation prone to error and inconsistency.	System automatically generates daily sales and basyo summary reports, viewable on the admin dashboard.
Performance Review	No way to generate summary or reports; owner relies on observation.	Difficulty in analyzing business performance and identifying sales trends, leading to reactive decision-making.	System provides automated Analytics and Trend Visualization (Top Products, Monthly Sales Cards, and Charts), enabling the owner to make data-driven decisions instantly.

PROCESS	CURRENT (AS-IS)	IDENTIFIED PROBLEMS	PROPOSED SYSTEM SOLUTION (TO-BE)
Collection & Delivery Routing	Staff conduct random, unsystematic sweeps of the neighborhood, checking houses one by one to see if they need refills.	Wasted manpower and time visiting customers who aren't empty, while missing those who actually need service.	System enables Strategic Routing by identifying customers with "aging" containers (e.g., 5 days since last refill), allowing for targeted and efficient visits.
Customer Profiling	Customer details are minimal (often just nicknames) and scattered across different logbooks without contact info.	Identity confusion (multiple customers with same name) and inability to contact customers for service updates.	System utilizes a Centralized Customer Database storing complete details (Name, Address, Contact #), ensuring accurate identification and history tracking.



**“Every great system
starts with a small idea —
and the courage to make it
real.”**



Thank You